

Reproduction of jiaweizzhao/GaLore: LLaMA-60M pre-training on C4 with GaLore

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Independent reproduction · Artifact evaluation · Machine learning

Abstract

Pre-trained LLaMA-60M on streamed C4 with the GaLore AdamW optimizer using the README command at the pinned commit (rank 128, scale 0.25, projection update every 200 steps, lr 0.01, bf16), for 2,000 steps instead of 10,000 to fit the time budget, with evaluation every 500 steps. Changes needed on this machine: micro-batch 64 with the same total batch of 512 (256 does not fit 16 GB), transformers 4.40.2 instead of the pinned 4.31 (no Python 3.12 build), and the validation set loaded as allenai/c4 instead of the retired c4 name. One NVIDIA RTX 5060 Ti.

About this record

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1 Sealed reports

The experiments were sealed with K-Veritas [1], which binds each result to the code, hardware and time that produced it.

Report	Label	Sealed	Runs	Invocations	Signer
r1 [2]	LLaMA-60M GaLore, 2000 st...	2026-10-10	1	1	Signed by the K-Veritas server

2 Results

r1: LLaMA-60M GaLore, 2000 steps

Run	Metric	Final value	Step
1	params	58.07	
1	enabled	25.3	
1	loss	3.8481	
1	eval_loss	3.8481	2000
1	eval_perplexity	46.9039	2000

Run	Inline claim	Value	Phase
1	final_eval_loss	3.8481	
1	final_eval_perplexity	46.9039	

3 Setup

Report	Item	Value
r1	Operating system	linux / amd64
r1	CPU	Intel(R) Xeon(R) W-2145 CPU @ 3.70GHz, 16 cores
r1	GPU	NVIDIA GeForce RTX 5060 Ti, 16311 MiB
r1	Memory	31.0 GB
r1	Total run time	4000.3 s

4 Files

Each report and code bundle must hash to these values. A bundle belongs to the report whose signed data names its hash.

File	SHA-256	Size
r1.pdf	0407c1dc5cb0ada57f34084dbe1ec5f9f88d2e25f01629853224fa373ac3e01d	6.8 MB
r1.kvbundle.zip	8fff306ee6827640b6a7389baa870efaea748ec278598d61b62383cecd2114f6	57 KB

5 Links and licence

Paper: <https://arxiv.org/abs/2403.03507>

Reproduction of: <https://github.com/jiaweizzhao/GaLore/tree/2cc66f88cce189e505affbb91042a8e77f5bf4e9>

Record: CC-BY 4.0. Code: Apache-2.0, as declared by the authors. Tags: galore, memory-efficient-training, llama, c4, pretraining.

Listed, not endorsed. Review checks the artifact, never the science. Reports are never edited; changing them makes a new version. Other fields are corrected in place, and every correction is logged in metadata.yaml.

References

[1] M. K. Keita and C. Homan. Computer Science Conferences Should Require Nonrepudiable Experimental Results. NeurIPS 2026 Position Paper Track. arXiv:2605.08586.

[2] Sealed experiment report r1 (LLaMA-60M GaLore, 2000 steps), kv:2610.00034v1/r1. Data hash 3ea1024baf8c7e0c27e846061055683df0b20293833dcb6e85c6acd0ddb29262.
<https://kveritas.org/records/2610.00034v1/r1.pdf>